Two new species of Polynemicola *Unnithan, 1971 (Monogenoidea: Microcotylidae) in Threadfin fishes (Perciformes: Polynemidae) from Visakhapatnam coast, Bay of Bengal, India*

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ORIGINAL ARTICLE

Two new species of *Polynemicola* Unnithan, 1971 (Monogenoidea: Microcotylidae) in Threadfin fishes (Perciformes: Polynemidae) from Visakhapatnam coast, Bay of Bengal, India

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Abstract Two new species of *Polynemicola* Unnithan, 1971 collected from gills of marine threadfin fishes *Polydactylus sextarius* and *Polydactylus plebeius* (Polynemidae) from Visakhapatnam coast, India are described. The genus *Polynemicola* is unique in possessing eversible spiny cirrus. The new species *Polynemicola sextariusii* was found to possess unique morpho-anatomical characters like median row of spines on the cirrus, 12 vaginal hooks, 5–7 testes in a single row. Another new species *Polynemicola glandularis* differs from existing species of *Polynemicola* in having prostatic gland cells conspicuously on either side of seminal vesicle, septate oral suckers and 20 vaginal hooks.

Keywords *Polynemicola* · *Polydactylus* · Prohaptor · Opisthaptor · Bilocuate

Introduction

Polynemids are epibenthic, found in the tropical and subtropical waters of all oceans. They constitute the most important commercial and recreational fisheries along the east coast of India due to their availability throughout the year. The commonly occurring polynemid fishes available from Andhra Pradesh coast are *Polydactylus sextarius* (Schneider), *P. sexfilis* (Valenciennes), *P. plebeius* (Broussonet), *Eleutheronema tetradactylum* (Shaw), *Filimanus heptadactyla* (Cuvier) and *Leptomelanosoma indicum* (Shaw). Some species also occur in the river mouths and estuaries. Polynemids being migratory fishes offer fair chance to harbour for a variety of parasitic infections. The monogenoids of the family *Polynemicola* frequently inhabit the gills of host fish. The characteristic feature of these parasites is an organ of attachment, opisthaptor consisting of clamps (Unnithan 1971). Among significant contributions on monogenoids of the group are the reports of Diesing (1858), Mac Callum (1917), Tripathi (1954, 1959), Unnithan (1971), Mamaev (1972, 1977a, b, 1986), Bravo-Hollis (1985), Jones and Gibson (1990), Hayward (1997), Garfias and Leon (1998), Zhang et al. (2001, 2003), Gudivada and Vankara (2010) and Hadi et al. (2019). It is interesting to note that Unnithan (1971) erected the genus *Polynemicola* for the species of Microcotylinae with eversible spiny cirrus, reported from polynemid fishes. 11 species were included under the genus *Polynemicola*, of which 7 were reported from polynemid fishes, 2 from Sciaenid fishes, 1 from Lutjanid fish, *Xenistius californiensis* and 1 from *Sillago sihama* which is an accidental host (Mamaev 1986; Bravo-Hollis 1985; Hayward 1997; Zhang et al 2001). According to World Register of Marine Species-WoRMS (2020), there are 14 species reported of which two species, *P. ambigua* and *P. californica* were designated as malformed suffix and are now accepted as *P. ambiguus* and *P. californicus*, another species *P. ichimidai* (Unnithan 1971) synonymised as *Bivagina tai* (Yamaguti 1938, 1963 and the new species *Polynemicola sextariusii* described in this paper is already registered (Gudivada and Vankara 2010) as nomen dubium which will be described in the present paper along with one more new species, *Polynemicola glandularis* from the gills of *Polydactylus*

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sextarius Bloch and Schneider, 1801 and *Polydactylus plebeius* Broussonet, 1782 recovered at Visakhapatnam coast, Andhra Pradesh, India. Genus and species identification was done with the aid of standard books of Yamaguti (1963), Gusev (1973, 1978) and Agarwal et al. (2008).

Materials and methods

A total of 159 dead specimens of *Polydactylus sextarius* and 77 dead specimens of *Polydactylus plebeius* were collected from the local fisherman and local fish markets near the Visakhapatnam coast (Latitude-17° 41′ 12.5340″ N, Longitude-83° 13′ 6.5388″ E), Bay of Bengal, Andhra Pradesh from June, 2005 to June, 2007. Gills were removed carefully and placed in a saline solution, then teased and contents were washed and observed. During the examination, 2 new species of monogenoids, 24 from *P. sextarius* and 12 from *P. plebeius* were recovered. Parasites were thoroughly washed and post-fixed in AFA (alcohol, Formaldehyde, acetic acid in 85:10:5) and stained with alum carmine. After differentiation with acid alcohol and brief washing they were dehydrated in alcoholic grades, cleared in Creosote and mounted in Canada balsam. Drawings were made with the aid of camera lucida and all measurements were expressed in millimeters. Microphotographs were taken and scale was given accordingly. Holotypes and paratypes of the parasites were preserved and stored in Museum of Zoology Department, Andhra University, Visakhapatnam (ZDAU) and the voucher specimens (one paratype each) from the new taxa are likely to be submitted to Zoological survey of India (ZSI), Hyderabad with a temporary accession number.

Results

Polynemicola sextariusii n.sp. (Figure 1, 1–2E)

Prevalence: 11.94%
Intensity: 1.26
Family: Microcotylidae Taschenberg, 1879
Genus: *Polynemicola* (Unnithan 1971)

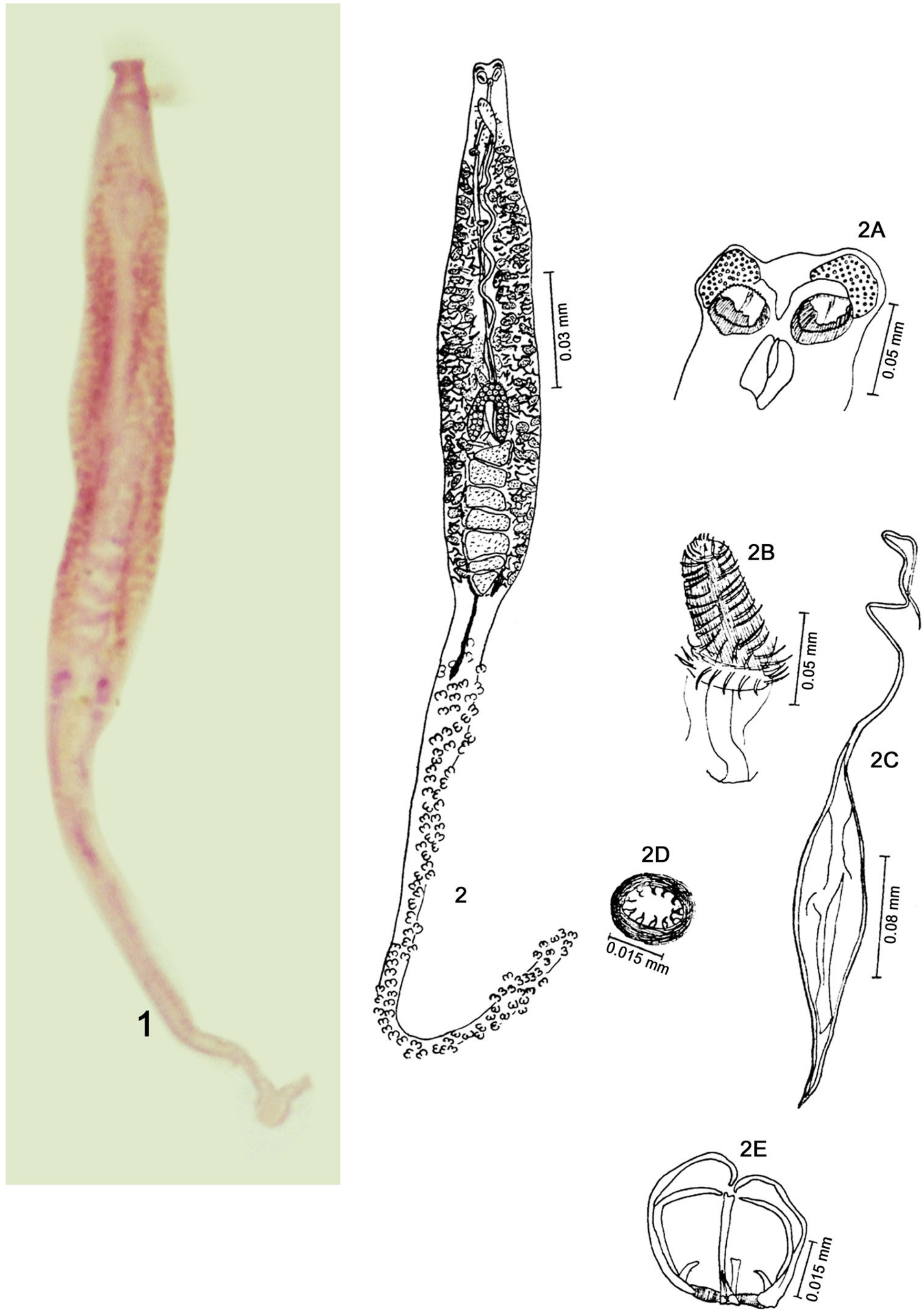
Description Body delicate, fragile, divisible into two regions, anterior body proper and posterior elongated opisthaptor which is more than the length of the body. Total length measures $1.13 - 1.70 \times 0.18 - 0.21$. Opisthaptor looks like a tail and measures 0.76–0.79 long, having numerous short stalked clamps on each side. At the base of the haptor the clamps are in 2–3 rows gradually become monoliform. Clamps are microcotylid type measuring $0.02 - 0.04 \times 0.02 - 0.05$, sclerites usually stout or angled. Median spring with two arms, dorsal and ventral. Distal part of ventral sclerite slightly grooved, dorsal sclerite reaches only half of ventral sclerite. Hinge line and lips of clamp capsule thick with fibrous tissue. Ventral wall of capsule thick with delicate fibrous tissue on each side of median spring. Conspicuous gland cells at the base of clamps indicate the origin of muscle fibres to each unit. Body proper elongate with anterior pointed and posterior broader region attached to the opisthaptor. Prohaptor possesses highly muscular oral suckers with cap like gland cells aggregated at the corners. Oral suckers spherical, aseptate, with a row of denticles on its rims and measures $0.01 - 0.02 \times 0.02 - 0.03$. Pharynx large, measuring $0.01 - 0.02 \times 0.01 - 0.02$. Oesophagus short with slight out pocketings and bifurcates into two crura just behind the genital atrium. Caeca ramify extensively on lateral sides than medially and extend slightly into the opisthaptor as simple caeca. Testes 5–7, subequal, variously shaped rectangular follicles arranged in a single row, post ovarian, occupying inter caecal field at a distance of 0.65–0.67 from the anterior end and open into vesicula seminalis. Eversible cirrus, highly muscular, cylindrical and measures $0.05 - 0.06 \times 0.03 - 0.04$. Cirrus bears short, sharp spines laterally and few more in the middle region. Genital atrium unarmed, muscular at a distance of 0.04–0.05 from the anterior end. Ovary single, long and inverted 'U' shaped. From the descending arm arises an oviduct just above the proximal lobe. Vitelline follicles co-extensive with the branches of caeca in external fields, not confluent posteriorily, do not enter opisthaptor. Vaginal pore spherical, mid dorsal, inner margin carries about twelve broad based, sharp triangular spines directed inwardly. Vaginal canal passing backwards and runs directly to ootype. Uterus ascending, pushing a median course into genital atrium, just anterior to male pore. Eggs elliptical, with bipolar filaments, one filament long and another short and measure $0.156 - 0.17 \times 0.05 - 0.06$.

Species: *Polynemicola sextariusii* n.sp.
Habitat. Gills
Name of the host: *Polydactylus sextarius* (Bloch and Schneider 1801)
No. of hosts infected. 19
No. of specimens. 24

Etymology The species name is named after the species name of the host.

Material examined *Holotype* (1). INDIA: Andhra Pradesh, Visakhapatnam coast, Andhra University Campus, Coll. Mani G, 16.ix.2006 with an accession number (Holotype- ZDAU-2006/CVL/GM/PS); *Paratypes* (22) with an accession number (Paratype- ZDAU-2006/CVL/

Fig. 1 1: Microphotograph of *Polynemicola sextaruisii* n.sp-40X. 2: *Polynemicola sextaruisii* n.sp. 2A: Prohaptor. 2B: Cirrus. 2C: Egg. 2D: Vagina. 2E: Clamp



GM/PS-I) submitted to Andhra University campus, Visakhapatnam coast, Andhra Pradesh, INDIA and one *paratype* (1) likely to submit to Zoological Survey of India (ZSI), Hyderabad with a temporary accession number (Paratype- ZSI-2006/CVL/GM/PS).

Discussion Unnithan (1971) established the genus *Polynemicola* for the species of Microcotylinae exceptionally with eversible spiny cirrus, with type taxon being *P. bulbovaginata* from *Polynemus indicus* from Indo-west pacific. Second species reported is *P. tritestis* from *Eleutheronema tetradactylum*. Unnithan (1971) also included *Microcotyle polynemi* (Mac Callum 1917; Tripathi 1954) under this genus. Mamaev (1986) listed seven described species (and one species inquirendum) known from the Indo-West Pacific (Arabian Sea, India, North Vietnam). He listed 6 species from polymenids perciforms and 2 from sciaenid fishes: *P. ambiguus* Mamaev (1977a, b); *P. polynemi* (Mac Callum 1917; Unnithan 1971); *P. bulbovaginata* (Unnithan 1971); *P. aequispinosa* (Mamaev 1971);

P.tritestis Unnithan 1971; *Polynemicola sp.inq.* (Syn. *Microcotyle* sp.) (Gupta and Khuller 1968) and two species, *P. heterocotyle* (Mamaev 1977a, b); *P. sciaenae* (Mamaev 1971) from sciaenids. Later, Bravo-Hollis (1985) reported *P. californicus* from Lutjanid fish, *Xenistuis californiensis*. Hayward (1997) reported *Polynemicola sp.* from *Sillago sihama* which is an accidental host. Later, Zhang et al. (2001) reported *P. brachypetala* from *Polynemus sextarius*. Recently, Hadi et al. (2019) described a new species, *Polynemicola indicus* from *Polynemus indicus* from Pakistan. So, till now, 11 valid species are described under this genus which is described in Table 1. The present parasites come under the genus *Polynemicola* because of eversible spiny cirrus and the host fish is *P. sextarius.* The present parasites differ from *P. indicus*, *P. tritestis*, *P. brachypetala* and *P. polynemi* in total length, number of testes, presence of median row of spines on the cirrus and the host difference. The present species resembles the type species, *P. bulbovaginata* in few characters like the general body shape, extension of caecae into opisthaptor and the presence of aggregated cephalic glands. But it differs from *P. bulbovaginata* in having median row of spines on the cirrus, 12 vaginal hooks, 5–7 testes in a single row and elongated opisthaptor. Though the new species resembles *P. indicus* in the presence of spiny cirrus and rightwardly directed haptor but it differs in many characters as above said. The present new species is compared with the existing species of *Polynemicola* in Table 2. Based on the above characters, it is felt justified to establish it as a new species and designated *Polynemicola sextariusii* after the species name of the host.

Polynemicola glandularis n.sp. (Figure 2, 1–2D)

Prevalence: 12.98%
Intensity: 1.2
Family: Microcotylidae Taschenberg, 1879
Genus: *Polynemicola* (Unnithan 1971)

Description Body elongate, divisible into an anterior flattened body and posterior elongated opisthaptor. Total length including opisthaptor measures $2.10-2.15 \times 0.23-0.26$. Body proper measures $1.15-1.18 \times 0.23-0.24$. Opisthaptor filamentous 0.94–0.97 in length, almost equal to the length of the body proper. Opisthaptor armed with numerous short, stalked clamps in two ventrolateral rows. Clamps similar in structure, but show slight variation in size, and measure $0.01-0.02 \times 0.02-0.03$. Clamps typically microcotylid with arcuate dorsal, ventral and basal sclerites enclosing the typical median spring. Body has head with highly scattered gland cells constituting the prohaptor. Oral suckers two, oval, oblique, bilocuate, measuring

Table 1 List of different species of *Polynemicola* from different hosts of different geographical locations

Sl. no.	Monogenean species	Host	Geographical locations	References
1	*P. ambiguus* (Mamaev 1977a, b)	Polynemid perciformes	Vladivostok, USSR	Mamaev (1977a, b)
2	*P. polynemi* (Mac Callum 1917; Unnithan 1971)	*E. tetradactylum, Leptomelanosoma indicum*	West coast, India	Unnithan (1971)
3	*P. bulbovaginata* (Unnithan 1971)	*Leptomelanosoma indicum*	West coast, India	Unnithan (1971)
4	*P. aequispinosa* (Mamaev 1971)	Polynemid perciformes	Vladivostok, USSR	Mamaev (1977a, b)
5	*P.tritestis* (Unnithan 1971)	*Eleuthreonema tetradactylum*	West coast, India	Unnithan (1971)
6	*P. heterocotyle* (Mamaev 1977a, b)	Percoid perciformes	Vladivostok, USSR	Mamaev (1977a, b)
7	*P. sciaenae* (Mamaev 1971)	Percoid perciformes	Vladivostok, USSR	Mamaev (1977a, b)
8	*P. californicus* (Bravo-Hollis 1986)	*Xenistuis californiensis*	Eastern pacific	Bravo-Hollis (1986)
9	*Polynemicola sp.* (Hayward 1997)	*Sillago sihama*	Indonesia	Hayward (1997)
10	*P. brachypetala* (Zhang et al. 2001)	*Polynemus sextarius*	East and South China Sea	Zhang et al. (2001)
11	*Polynemicola indicus* (Hadi et al. 2019)	*Polynemus indicus*	Karachi coast, Pakistan	Hadi et al. (2019)
12	*Polynemicola sextariusii* n.sp	*Polydactylus sextarius*	Visakhapatnam coast, India	Present study
13	*Polynemicola glandularis* n.sp	*Polydactylus plebeius*	Visakhapatnam coast, India	Present study

Table 2 Comparison of *Polynemicola sextariusii* n.sp with the other related species of *Polynemicola*

Characters	*P. bulbovaginata* (Unnithan 1971)	*P. tritestis* (Unnithan 1971)	*P. polynemi* (Unnithan 1971)	*P.indicus* (Hadi et al. 2019)	*P. sextariusii* n.sp.
Host	*Leptomelanosoma indicum*	*Eleuthreonema tetradactylum*	*E. tetradactylum, L. indicum*	*Polynemus indicus*	*Polydactylus sextarius*
Number of testes	5–7 pairs in two rows	3 in number	40 in number	6 in number	5–7 in a single row
Median row of spines on cirrus	Absent	Absent	Absent	Present	Present
No. of vaginal hooks	Numerous	20	Numerous	Not mentioned	12
Opisthaptor	Small	Small	Large	Large	Large

$0.01 - 0.03 \times 0.04 - 0.05$ with a row of denticles on the rims and thick walled. Mouth slit like, pharynx elongated, slightly overlapping the oral suckers, ornamented with a row of denticles and measures 0.02–0.03 long. Oesophagus short, and bifurcates into two crura just behind the cirrus. Crura ramify extensively on median and lateral sides, and terminating at the base of the opisthaptor at different levels. Testes 5–7 in number, rectangular, post ovarian, intercaecal at a distance of 0.89–0.92 from the anterior end. Vas deferens spherical, opens into vesicula seminalis. Prostatic gland cells lie on either side of vesicula seminalis. Ejaculatory duct anterior, narrow coiled tube, opening at the base of cirrus above the genital atrium. Cirrus long, eversible, highly muscular and measures 0.04–0.06. Cirrus bears lateral rows of long spines and irregularly arranged spines medially; base is wide bearing spines in 2–3 rows. Genital atrium unarmed. Ovary single, horse-shoe shaped, at a distance of 0.78–0.80 from the anterior end. Vitellaria follicular, co-extensive with branches of caeca but not entering into opisthaptor. Vitelline reservoir Y-shaped with two lateral arms joining anterior to the level of ovary. Vaginal pore circular, sometimes elongated transversely at a distance of 0.25–0.26 from anterior end, carrying more than 20 broad based sharp, triangular spines directed inwardly. Vagina measures 0.04 in diameter. Uterus sinuous, opens into genital atrium. Eggs bipolar.

Species: *Polynemicola glandularis* n.sp.
Habitat. Gills
Name of the host. *Polydactylus plebeius* Broussonet, 1782
No. of hosts infected. 10
No. of specimens. 12

Etymology The species name is named on the basis of presence of prostatic gland cells on either side of vesicula seminalis.

Material examined *Holotype* (1). INDIA: Andhra Pradesh, Visakhapatnam coast, Andhra University Campus, Coll. Mani G, 25.xi.2006, with an accession number (Holotype- ZDAU-2006/CVL/GM/PG); *Paratypes* (10) with an accession number (Paratype- ZDAU-2006/CVL/GM/PG-I), submitted to Andhra University campus, Visakhapatnam coast, Andhra Pradesh, India and one *paratype* (1) likely to submit to Zoological Survey of India (ZSI), Hyderabad with a temporary accession number (Paratype- ZSI-2006/CVL/GM/PG).

Discussion Unnithan (1971) included three species under the genus *Polynemicola*, *P. bulbovaginata, P. tritestis and P. polynemi.* Hadi et al. (2019) also described a new species, *P.indicus* from *polynemus indicus* of Pakistan. In the present work, two new species were compared with the valid species in the genus. Present parasites differ from *P. polynemi*, *P.indicus* and *P. tritestis* in having a wide difference in the number of testes and also other characters. Present parasites differ from the type-species *P. bulbovaginata* in having septate buccal suckers, denticulate oral suckers, pharynx, 5–7 testes in a single row, circular and transversely elongated vagina and the host species. Present parasites resemble *P. sextariusii* n.sp. in having 5–7 number of rectangular shaped testes. But they differ from *P. sextariusii* n.sp. in general shape of the body, highly scattered gland cells in the head region, length of opisthaptor almost equal to the length of body proper, having prostatic gland cells conspicuously on either side of seminal vesicle, 20 vaginal spines and different host i.e. *P. plebeius.* A comparative table is given for comparing the present new species with the closely related species of *Polynemicola* (Table 3). Based on the above differences, it was felt apt to designate them to the level of species, *Polynemicola glandularis* n.sp.

Fig. 2 1: Microphotograph of *Polynemicola glandularis* n.sp-40X. 2: *Polynemicola glandularis* n.sp. 2A: Prohaptor. 2B: Cirrus. 2C: Vagina. 2D: Clamp

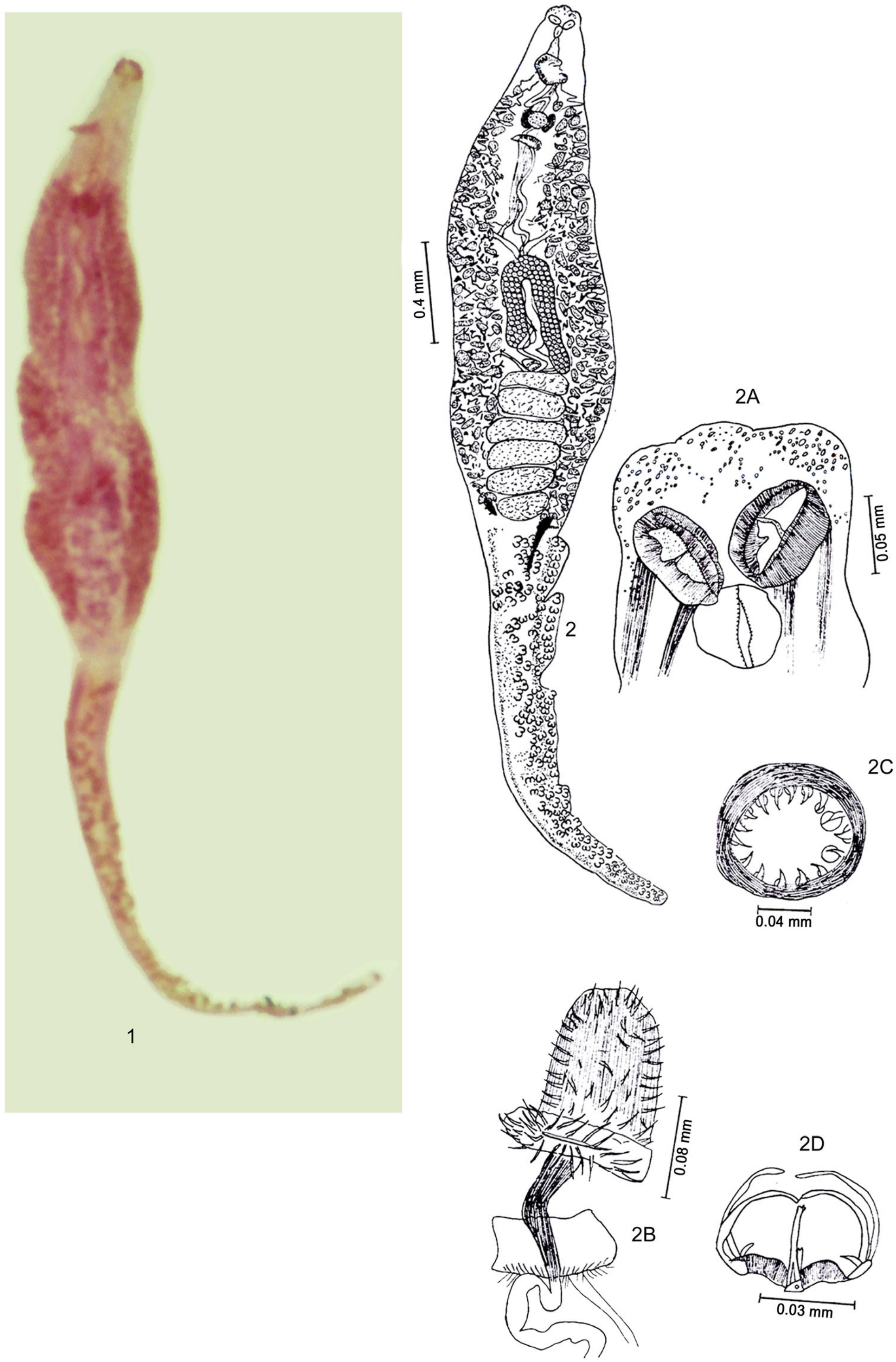


Table 3 Comparison of *Polynemicola glandularis* n.sp with the other related species of *Polynemicola*

Characters	*P.bulbovaginata* (Unnithan 1971)	*P.tritestis* (Unnithan 1971)	*P.polynemi* (Unnithan 1971)	*P.indicus* (Hadi et al. 2019)	*P.sextariusii* n.sp.	*P.glandularis* n.sp.
Host	*Leptomelanosoma indicum*	*Eleuthreonema tetradactylum*	*E. tetradactylum, L. indicum*	*Polynemus indicus*	*Polydactylus sextarius*	*Polydactylus plebeius*
Oral suckers	Aseptate	Aseptate	Aseptate	2 in number, not mentioned	Aseptate	Septate
Number of testes	5–7 pairs in two rows	3 in number	40 in number	6 in number	5–7 in a single row	5–7 in a single row
Prostatic gland cells on either side of seminal vescicle	Absent	Absent	Absent	Not mentioned	Absent	Present
Median row of spines on cirrus	Absent	Absent	Absent	Present, 30	Present	Present
No. of vaginal hooks	Numerous	20	Numerous		12	20

Conclusion

The present study allowed identification of two species of microcotylid monogenoids from polynemid fishes from Viskhapatnam coast, Bay of Bengal, Andhra Pradesh which is new to this geographical area.

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Author's contribution The first author was involved in collecting the fish samples and parasites, literature collection and the second author helped in writing and drafting the manuscript.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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